

Quiz: Embeddings & Vector Databases

Answered Version

Section A: Multiple Choice Questions (MCQs)

1. What is an embedding in AI?

- a) A database table
- b) A numerical representation of text, images, or data
- c) A programming language
- d) A type of neural network

Answer: b) A numerical representation of text, images, or data

2. Why are embeddings useful in AI applications?

- a) They compress files
- b) They convert data into numbers that machines can compare
- c) They increase internet speed
- d) They replace databases

Answer: b) They convert data into numbers that machines can compare

3. Which technique is commonly used to measure similarity between embeddings?

- a) Binary search
- b) Cosine similarity
- c) Linear regression
- d) Sorting algorithm

Answer: b) Cosine similarity

4. What is the main purpose of a vector database?

- a) Store only text documents
- b) Store and search numerical vector representations efficiently
- c) Store images only
- d) Replace relational databases completely

Answer: b) Store and search numerical vector representations efficiently

5. Which of the following is a popular vector database?

- a) MySQL
- b) Pinecone
- c) Excel
- d) Oracle Forms

Answer: b) Pinecone

Section B: True / False

6. Embeddings convert text or images into numerical vectors.

Answer: True

7. Vector databases are optimized for similarity search.

Answer: True

8. Cosine similarity measures how similar two vectors are in direction.

Answer: True

9. Traditional SQL databases are always faster than vector databases for similarity search.

Answer: False

Section C: Short Answer

10. In simple words, what is similarity search?

Answer: Similarity search means finding the items that are most similar to a given query based on meaning or closeness in vector space, not just exact keyword matching.

11. Give one real-world application of embeddings.

Answer: One real-world application is a recommendation system, where embeddings help suggest similar products, movies, or songs based on user preferences.

12. What happens when two embeddings are very close in vector space?

Answer: It usually means the two pieces of data are semantically very similar or related in meaning.

Section D: Practical Thinking Questions

13. Scenario: You build a chatbot for a university website. The chatbot should answer questions from course PDFs and policy documents. Question: Why would embeddings + a vector database be useful here?

Answer: Embeddings convert the text from PDFs and policy documents into vectors, and the vector database stores them for fast semantic search. This helps the chatbot retrieve the most relevant passages even when the student asks the question in different words.

14. Scenario: A user searches for "cheap smartphones", but the database contains products labeled "budget mobile phones." Question: How can embeddings help retrieve the correct results even if the exact words are different?

Answer: Embeddings capture semantic meaning, so "cheap smartphones" and "budget mobile phones" end up close in vector space. Because of that, the system can retrieve relevant products even without exact keyword overlap.

Section E: Mini Hands-On Question

15. Steps Challenge: Arrange the steps for building a semantic search system in the correct order: a) Store embeddings in a vector database b) Convert documents into embeddings c) Convert the user query into an embedding d) Find the most similar vectors

Answer: Correct order: b) Convert documents into embeddings -> a) Store embeddings in a vector database -> c) Convert the user query into an embedding -> d) Find the most similar vectors